

0.0.1 Interacting Multipoles of Rank 2 and 4

[illegible]

Simplify deltas:

[illegible]

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315}.$$

[illegible]

(2b)

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\epsilon\zeta}^A \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\delta\epsilon}^A \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\epsilon}^A \\
& -3780 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& -3780 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\gamma\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
& +1260 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
& +9 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A + 36 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (2d)
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& + \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\delta\delta}^A \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{zzz\beta}^A \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& +1260 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +9 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A + 36 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& -1890 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\delta\delta}^A - 2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -3780 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{zzz\beta}^A - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\delta\delta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +9 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A + 36 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -1890 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A - 2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \\
& +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +9 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A + 36 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3c)
\end{aligned}$$

$$\begin{aligned}
&\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
&+10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 5670 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
&-7560 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
&+315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\
&+1260 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
&+1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A + 9 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
&+36 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3d)
\end{aligned}$$

Multiply Out Sum:

[illegible]

[illegible]

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned} &+990 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 1836 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{xzzz}^A \\ &- 1836 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{yyzz}^A - 306 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{zzzz}^A \\ &- 306 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{zzzz}^A + 324 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A \\ &+ 648 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A + 324 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{yyyy}^A \\ &+ 1944 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{xzzz}^A + 648 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A \\ &+ 648 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{xzzz}^A + 1944 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \\ &+ 45 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A + 54 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A \\ &+ 9 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A + 9 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A \\ &+ 54 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A + 45 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A \end{aligned} \right] \quad (4c)$$

Combining everything:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = \left[\begin{aligned} & + \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxzz}^A \\ & - \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyzz}^A - \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{zzzz}^A \\ & - \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{zzzz}^A + \frac{36}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A \\ & + \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A + \frac{36}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyyy}^A \\ & + \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A + \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A \\ & + \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxzz}^A + \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \\ & + \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A + \frac{6}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A \\ & + \frac{1}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A + \frac{1}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A \\ & + \frac{6}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A + \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A \end{aligned} \right] \quad (5)$$

A:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = \left[\begin{aligned} & -\frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A - \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyzz}^A \\ & + \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A + \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A \\ & + \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A + \frac{204}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyyy}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxzz}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \\ & + \frac{36}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A + \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A \\ & + \frac{36}{35} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyyy}^A - \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A \\ & - \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A - \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A \\ & - \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A - \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A \\ & - \frac{72}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A - \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A \\ & - \frac{216}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A + \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A \\ & + \frac{6}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A + \frac{1}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A \\ & + \frac{1}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A + \frac{6}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A \\ & + \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} \left[\begin{aligned} & -\frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxz}^A - \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyzz}^A \\ & + \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxxx}^A + \frac{96}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{xxyy}^A \\ & + \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{yyyy}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxzz}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A - \frac{211}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A \\ & - \frac{282}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A - \frac{71}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A \\ & - \frac{71}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A - \frac{282}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A \\ & \quad - \frac{211}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A \end{aligned} \right] \quad (6b)$$

B:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = \left[\begin{aligned} & + \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A + \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxzz}^A \\ & + \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A + \frac{22}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \\ & - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A \\ & - \frac{96}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A - \frac{96}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A \\ & - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{axzz}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{ayzz}^A \\ & + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{axzz}^A + \frac{34}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{ayzz}^A \\ & - \frac{211}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{axxx}^A - \frac{282}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{axyx}^A \\ & - \frac{71}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{ayyy}^A - \frac{71}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{axxx}^A \\ & - \frac{282}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{axxy}^A - \frac{211}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{ayyy}^A \end{aligned} \right] \quad (7a)$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} \left[\begin{aligned}
& + \frac{144}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A + \frac{144}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxzz}^A \\
& + \frac{144}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyzz}^A + \frac{144}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \\
& - \frac{451}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxxx}^A - \frac{311}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxxx}^A \\
& - \frac{762}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxyy}^A - \frac{762}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{xxyy}^A \\
& - \frac{311}{35} \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{yyyy}^A - \frac{451}{35} \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyyy}^A
\end{aligned} \right] \tag{7b}
\end{aligned}$$